

ECON 311

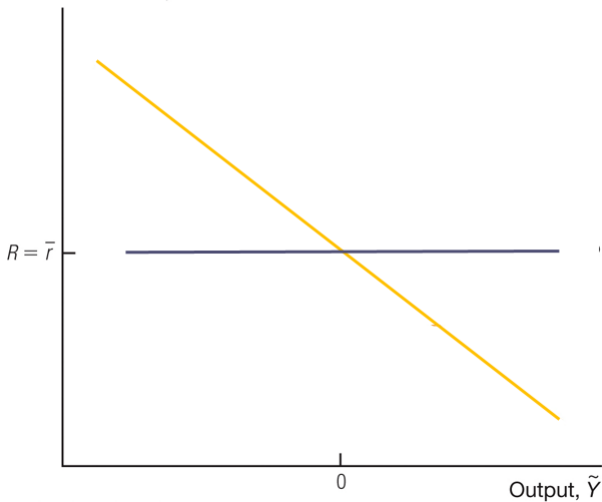
Lecture 15: The IS-MP Model, and the Phillips Curve

The IS-MP model

- The MP curve
 - ▶ Illustrates the central bank's ability to set the real interest rate at a particular value (horizontal line)
- The IS curve
 - ▶ Illustrates the negative relationship between short-run output and interest rates (through changes in aggregate expenditure)

The IS-MP model

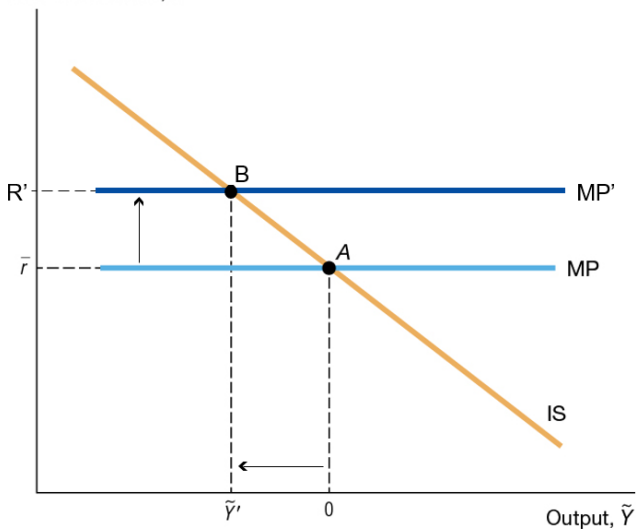
The MP Curve in the IS-MP Diagram
Real interest rate, R



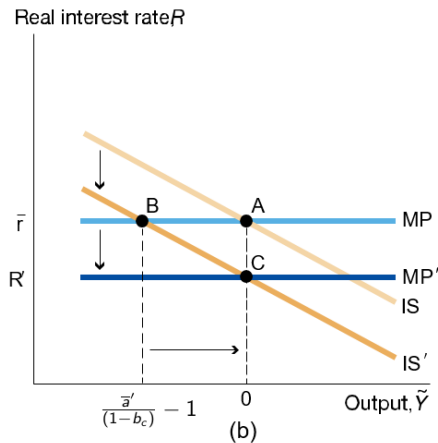
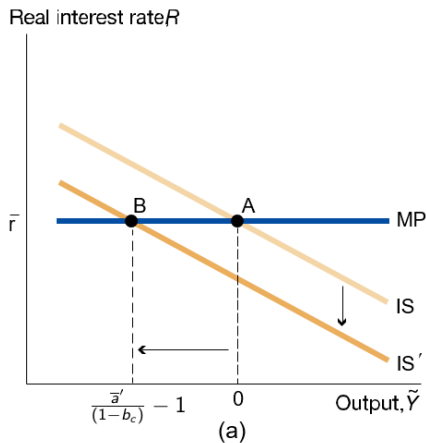
The IS-MP model

Raising the Interest Rate in the IS-MP Diagram

Real interest rate, R



Temporary Negative Demand Shock



IS-MP Model

- In this simple model, the CB can avoid recessions or overheating economies through active monetary policy.
 - ▶ However, in practice there is usually a lagged between shocks and the response of the CB due to two factors:
 1. It usually takes time for policy makers to understand the extent of the shocks.
 2. It takes time for the policy to affect the economy (between 6-18 months for changes in the interest rate to have substantial effect on economy activity).
 - ▶ Business Cycles are the result of the interaction of shocks and the response of the CB and governments to these shocks.

The Phillips Curve

IS-MP Model and the Phillips Curve

The Phillips Curve

- Inflation is defined as:

$$\pi_t = \frac{P_{t+1} - P_t}{P_t}$$

- Firms set their prices on the basis of:
 - ▶ Their expectations of the economy-wide inflation rate
 - ▶ The state of demand for their product

The Phillips Curve

- Inflation is then determined by the following equation:

$$\pi_t = \pi_t^e + \bar{v}\tilde{Y}_t$$

- We are going to assume: $\pi_t^e = \pi_{t-1}$
 - ▶ Firms adjust their forecast slowly.
 - ▶ This embodies the sticky inflation assumption

The Phillips Curve

- Using the equations:

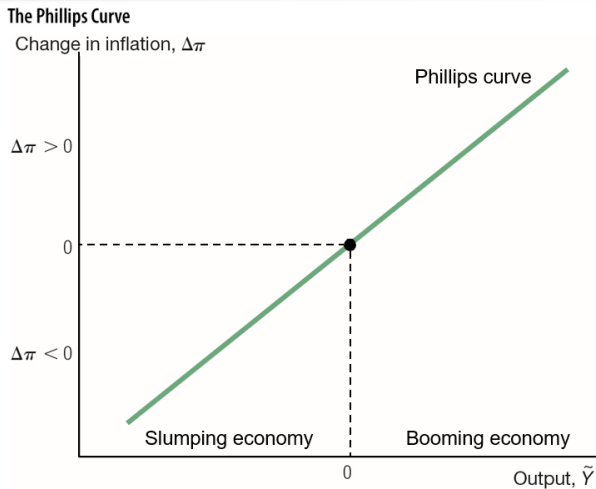
$$\pi_t = \pi_{t-1} + \bar{v}\tilde{Y}_t$$

$$\Delta\pi_t = \pi_t - \pi_{t-1}$$

- Therefore, the Phillips curve can be expressed as:

$$\Delta\pi_t = \bar{v}\tilde{Y}_t$$

The Phillips Curve

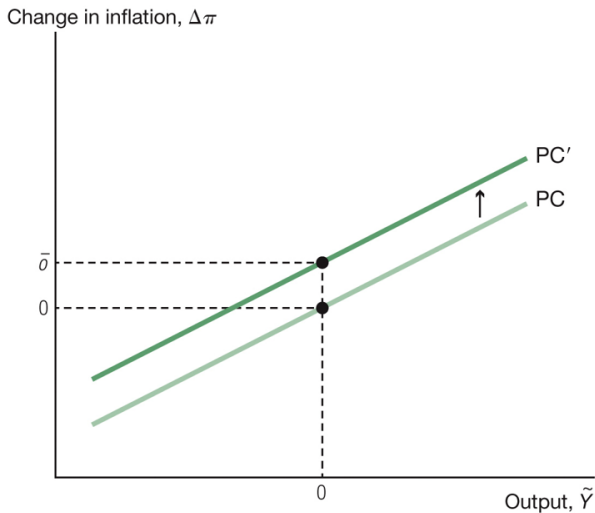


Inflation Shocks

- In our IS-MP model, we are going to also include inflation shocks (called cost-push inflation)
 - ▶ We are going to include these shocks in our Phillips curve as follows:

$$\Delta\pi_t = \bar{v}\tilde{Y}_t + \bar{o}$$

The Phillips Curve with cost-push inflation



Sticky Inflation

- The sticky inflation assumption is a very important property of this model
 - ▶ Inflation displays inertia so any change to prices in the present affects inflation in the future
 - ▶ Shocks to the Phillips Curve have long memory
 - ▶ A one time economic policy affects inflation rates in multiple periods

The Phillips Curve

IS-MP Model and the Phillips Curve

Cost-Push Inflation and Monetary Policy

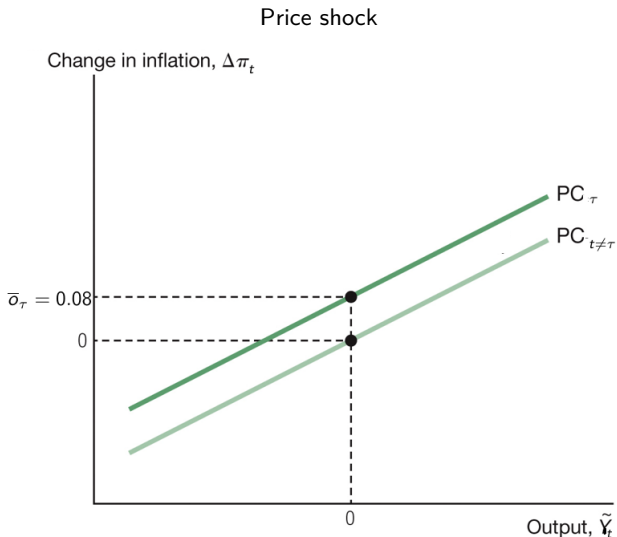
- Suppose that the economy is at an equilibrium with 2% inflation and that at $t = \tau$, there is a one time increase in energy prices due to OPEC raising the price of oil that increases inflation to 10%.

$$\Delta\pi_{t \neq \tau} = \underbrace{\bar{v}}_0 \widetilde{Y}_t$$

$$\Delta\pi_{\tau} = \underbrace{\bar{v}}_0 \widetilde{Y}_t + \underbrace{\bar{o}_{\tau}}_{0.08}$$

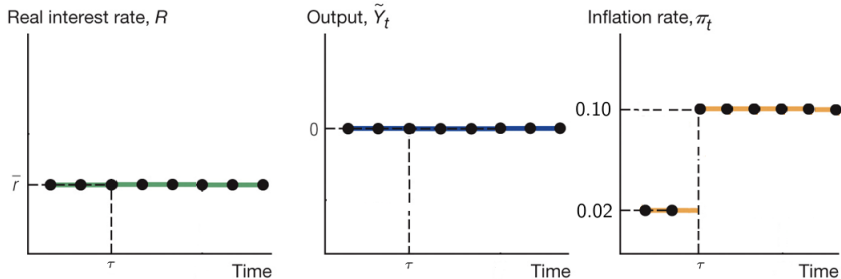
- In the absence of any other changes, this shock keeps inflation at a higher level for all subsequent periods.

Cost-Push Inflation and Monetary Policy



Cost-Push Inflation and Monetary Policy

Dynamics of price shock

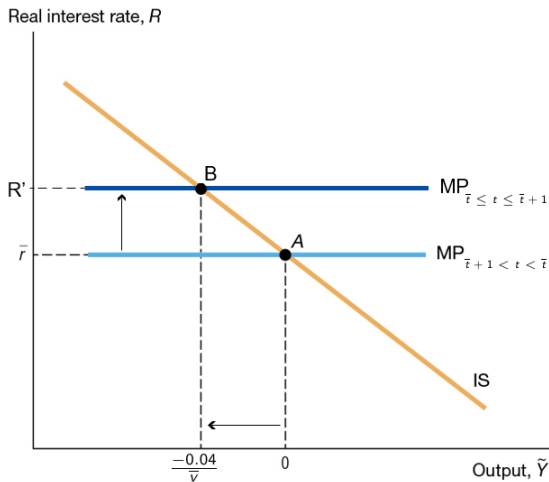


Cost-Push Inflation and Monetary Policy

- Given the relative high level of inflation, the government decides, at time $t = \bar{t} > \tau$, to use monetary policy to bring down inflation.
- Rather than creating a large decrease in output, the government usually takes several periods to reduce inflation.
 - ▶ We will assume here that the government policy reduces inflation at 4% per period.
 - ▶ In the absence of any other shock, it takes the economy 2 periods to return to the initial inflation level.

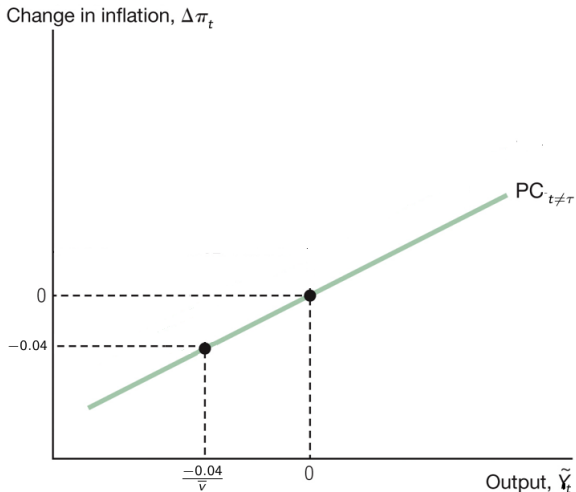
Cost-Push Inflation and Monetary Policy

Monetary Policy



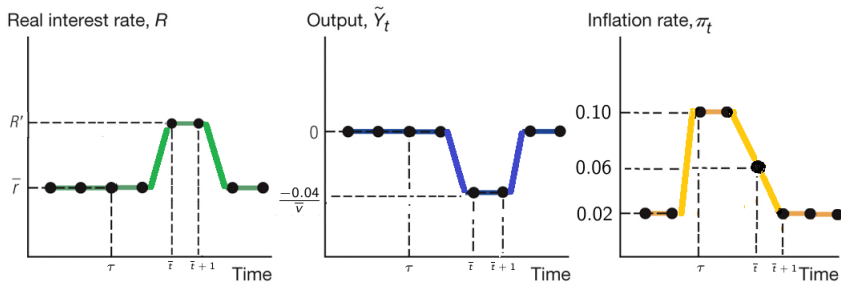
Cost-Push Inflation and Monetary Policy

Monetary Policy and inflation



Cost-Push Inflation and Monetary Policy

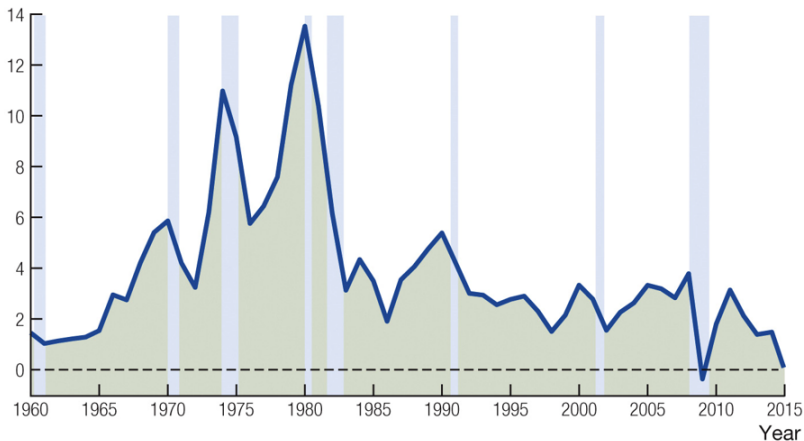
Dynamics of Monetary Policy



Cost-Push Inflation and Monetary Policy

Inflation in the United States

Percent



Practice Question: Demand Shock

Suppose that an economy similar to the one studied in class is at its potential output equilibrium with a constant 1% inflation rate. Using the full short-run model (IS-MP and Phillips curve), explain what happens to the economy over time if there is a demand shock that lasts for two periods. Be sure to include graphs showing how output and inflation respond over time.